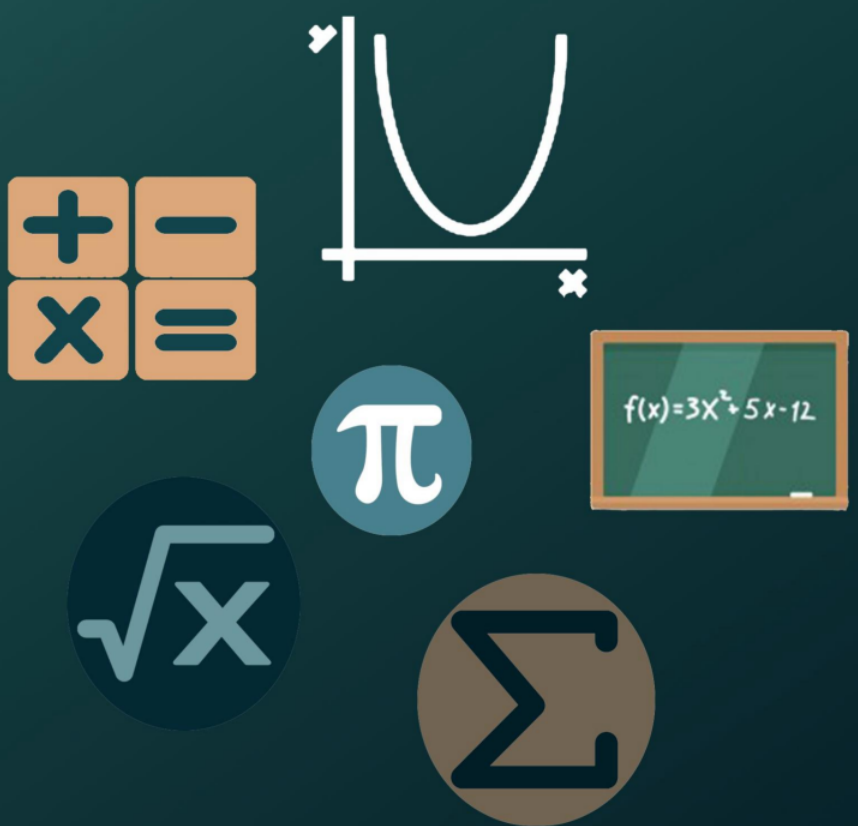
MATHEMATICS

Probability



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Basic Definitions related to Probabitliy				
2	Classical Definition of Probability	1-8	16-20	1-2	12-14,18,21-23
3	Odds in Favour and Odds in Against	9			24
4	Algebra of Probability	10	1-2,21		
5	Addition Theorem of Probability	11-16	3,14,22		1
6	Conditional Probability	17,18	23	3-4	
7	Multiplication Theorem of Probability	19,20	24,25	5-6	25
8	Dependent and independent Events	21,22	4,26-28	7-12	2-5
9	Probability Associated with Three Events		5,29	13	
10	Total Probability Theorem	23-29	8-13,30-32	14-17	6-8
11	Bayes' Theorem	30	15	18-21	9,19,26,27
12	Probabilities Through Stochastic Tree Diagram			22	10-11
13	Mathematical Expectation				20
14	Geometrical Probability			23-24	
15	Probability Distribution	31			
16	Bionomial Probability Distribution	32-34	6,7		
17	After Complete Chapter		33		15-17,28,29

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Classical Definition of Probability

Q1. The probability that a leap year selected at random will contain 53 Sundays, is :

(A) $\frac{5}{7}$ (B) $\frac{2}{7}$

(C) $\frac{1}{7}$ (D) $\frac{4}{7}$

Q2. Three coins are tossed all together. The probability of getting at least two heads is :

(A) $\frac{1}{8}$ (B) $\frac{3}{8}$

(C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Q3. The probability, that a non-leap year has 53 Sundays is :

(A) $\frac{2}{7}$ (B) $\frac{3}{7}$

(C) $\frac{4}{7}$ (D) $\frac{1}{7}$

Q4. A bag contains 5 white, 7 black and 4 red balls. Three balls are drawn from the bag at random. The probability that all the three balls are white, is :

(A) $\frac{1}{56}$ (B) $\frac{3}{5}$

(C) $\frac{1}{60}$ (D) $\frac{3}{16}$

Q5. Out of 40 consecutive natural numbers, two are chosen at radnom. Probability that the sum of the numbers is odd, is :

(A) $\frac{14}{29}$ (B) $\frac{20}{39}$

(C) $\frac{1}{2}$ (D) None of these

Q6. There are four letters and four addressed envelopes. The chance that all letters are not dispatched in the right envelope, is :

(A) $\frac{19}{24}$ (B) $\frac{23}{24}$

(C) $\frac{21}{23}$ (D) None of these

Q7. A pair of dice is thrown, if 5 appears on at least one of the dice, then the probability that the sum is 10 or greater is :

(A) $\frac{1}{12}$ (B) $\frac{11}{36}$

(C) $\frac{2}{9}$ (D) $\frac{3}{11}$

Q8. Two cards are drawn at random from a pack of 52 cards. The probability that both of them are king is-

(A) $\frac{2}{13}$ (B) $\frac{1}{169}$

(C) $\frac{1}{221}$ (D) $\frac{30}{221}$

Odds in Favour and Odds in Against

Q9. n persons sit on a round table. The odd against two specified persons sitting together is :

(A) $2 : (n-3)$ (B) $(n-1) : 2$

(C) $(n-2) : 2$ (D) $(n-3) : 2$

Algebra of Probability

Q10. A bag contains 5 brown and 4 white socks. A man pulls out two socks. The probability that these are of the same colour, is :

(A) $\frac{18}{308}$ (B) $\frac{30}{108}$

(C) $\frac{48}{108}$ (D) $\frac{5}{108}$

Addition Theorem of Probability

Q11. Two events A and B have probabilities 0.25 and 0.50 respectivley. The probability that both A and B occur simultaneously is 0.14, then probability that neither A nor B occurs is :

(A) 0.39 (B) 0.11

(C) 0.25 (D) 0.11

Q12. One card is drawn from a pack of 52 cards. The probability that it is a king or diamond is :

(A) $\frac{3}{13}$ (B) $\frac{3}{26}$

(C) $\frac{1}{26}$ (D) $\frac{4}{13}$

Q13. The probability that at least one of the events A and B occurs is 0.6.

If A and B occurs simultaneosuly with probability 0.2, then $P(\overline{A})+P(\overline{B})$ is equal to :

(A) 1.4 (B) 1.2

(C) 0.4 (D) 0.8

Q14. A box contains 6 nails and 10 nuts. Half of the nails and half of the nuts are rusted. If one item is chosen at random, the probability that it is rusted or is a nail, is :

(A) $\frac{11}{16}$ (B) $\frac{14}{16}$

(C) $\frac{5}{16}$ (D) $\frac{3}{16}$

Q15. If A and B are any two events, then $P(\overline{A}\cap B)$ is

(A) P(B) – P(A ∩ B)

(B) $P(A)\cdot P(\overline{B})$

(C) 1 – P(A) – P(B)

(D) $P(\overline{A})+P(\overline{B})-P(\overline{A}\cap\overline{B})$

Q16. A die is thrown. Let A be the event that the number obtained is greater than 3.

Let B be the event that the number obtained is less than 5. Then $P(A\cup B)$ is

(A) $\frac{3}{5}$ (B) 0 (C) 1 (D) $\frac{2}{5}$

Conditional Probability

Q17. From a pack of 52 cards, two cards are drawn one by one without replacement. The probability that first drawn card is a king and second is queen, is

(A) $\frac{4}{663}$ (B) $\frac{2}{13}$

(C) $\frac{8}{663}$ (D) $\frac{103}{663}$

Q18. It is given that the events A and B are such that $P(A)=\frac{1}{4}$, $P(\frac{A}{B})=\frac{1}{2}$ and $P(\frac{B}{A})=\frac{2}{3}$.

Then P(B) is

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Multiplication Theorem of Probability

Q19. Two cards are drawn one by one at random from a pack of 52 cards without replacement. The probability of those two being aces is :

(A) $\frac{1}{26}$ (B) $\frac{1}{221}$

(C) $\frac{1}{2}$ (D) None of these

Q20. A coin is tossed three times in succession. If E is the event that there are at least two heads and F is the event in which first throw is a head, then P(E/F) is equal to :

(A) $\frac{3}{4}$ (B) $\frac{3}{8}$

(C) $\frac{1}{2}$ (D) $\frac{1}{8}$

Dependent and independent Events

Q21. A coin is tossed and a dice is rolled. The probability that the coin shows the head and the die shows 6 is :

(A) $\frac{1}{12}$ (B) $\frac{1}{24}$

(C) $\frac{1}{2}$ (D) $\frac{1}{6}$

Q22. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is :

(A) $\frac{7}{9}$ (B) $\frac{8}{9}$

(C) $\frac{1}{9}$ (D) $\frac{2}{9}$

Total Probability Theorem

Q23. The items produced by a firm are supposed to contain 5% defective items. The probability that a sample of 8 items will contain less than 2 defective items is :

(A) $\frac{35}{16}\left(\frac{1}{20}\right)^6$ (B) $\frac{153}{20}\left(\frac{1}{20}\right)^7$

(C) $\frac{27}{20}\left(\frac{19}{20}\right)^7$ (D) $\frac{533}{400}\left(\frac{19}{20}\right)^7$

Q24. A bag contains 3 black and 4 white balls. Two balls are drawn one by one at random without replacement. The probability that the second drawn ball is white is :

(A) $\frac{4}{7}$ (B) $\frac{1}{7}$

(C) $\frac{4}{49}$ (D) $\frac{12}{49}$

Q25. In a box there are 2 red, 3 black and 4 white balls. Out of these balls are drawn together. The probability of these being of same colour is :

(A) $\frac{5}{84}$ (B) $\frac{1}{21}$

(C) $\frac{1}{84}$ (D) None of these

Q26. If A speaks truth in 75% cases and B in 80% cases, then the probability that they contradict each other in during the same statement, is :

(A) $\frac{13}{20}$ (B) $\frac{12}{20}$

(C) $\frac{7}{20}$ (D) $\frac{2}{5}$

Q27. Assuming that for a husband-wife couple the chances of their child begining a boy or a girl are the same, the probability of their two children being a boy and a girl is :

(A) $\frac{1}{8}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 1

Q28. A coin is tossed 4 times. The probability that at least one head turns up is :

(A) $\frac{14}{16}$ (B) $\frac{1}{16}$

(C) $\frac{15}{16}$ (D) $\frac{2}{16}$

Q29. Five coins whose faces are marked 2, 3 are tossed. The chance of obtaining a total of 12 is :

(A) $\frac{1}{16}$ (B) $\frac{1}{32}$ (C) $\frac{5}{16}$ (D) $\frac{3}{16}$

Bayes' Theorem

Q30. In a college, 25% of the boys and 10% of the girls offer Mathematics. The girls constitute 60% of the total number of students. If a student is selected at random and is found to be studying Mathematics, the probability that the student is a girl, is :

(A) $\frac{5}{8}$ (B) $\frac{1}{6}$ (C) $\frac{3}{8}$ (D) $\frac{5}{6}$

Probability Distribution

Q31. The mean and variance of a binomial distribution are 4 and 3 respectivley, then the probability of getting exactly six successes in this distribtuion is :

(A) ${}^{16}C_6\left(\frac{1}{4}\right)^{10}\left(\frac{3}{4}\right)^{6}$ (B) ${}^{16}C_6\left(\frac{1}{4}\right)^{6}\left(\frac{3}{4}\right)^{10}$

(C) ${}^{12}C_6\left(\frac{1}{4}\right)^{10}\left(\frac{3}{4}\right)^{6}$ (D) ${}^{12}C_6\left(\frac{1}{4}\right)^{6}\left(\frac{3}{4}\right)^{6}$

Bionomial Probability Distribution

Q32. 8 coins are tossed at a time. The probability of getting at least 6 heads up is :

(A) $\frac{37}{256}$ (B) $\frac{9}{64}$

(C) $\frac{229}{256}$ (D) $\frac{57}{64}$

Q33. In tossing 10 coins the probability of getting exactly 5 heads is :

(A) $\frac{1}{2}$ (B) $\frac{63}{256}$

(C) $\frac{193}{256}$ (D) $\frac{9}{128}$

Q34. One hundred identical coins each with probability p, of showing up heads are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins at equal to that of heads showing on 51 cons, then the value of p is :

(A) $\frac{51}{101}$ (B) $\frac{1}{21}$

(C) $\frac{49}{101}$ (D) $\frac{50}{101}$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two fair dice are thrown simultaneously. 3 events A, B & C are defined as follows :

A : odd face on first dice

B : odd face with second dice

C : sum of the points on two dice is odd.

Then :

(A) the events A, B & C are mutually exclusive

(B) the events A, B & C are mutually independent

(C) the events A, B & C are pairwise independent

(D) $P(A \cap B) = P(B \cap C) = P(C \cap A)$.

Q2. E and F are independent events such that $0 < P(E), P(F) < 1$, then

(A) E and F are mutually exclusive

(B) E and $\overline{F}$ are mutually independent

(C) $\overline{E}$ and $\overline{F}$ are mutually independent

(D) $P\left(\frac{E}{F}\right)+P\left(\frac{\overline{E}}{F}\right)=1$

Q3. A and B are two among 10 students from which 3 students are to be selected at random to form a committee. Then,

(A) Probability that A belongs to the committee is $\frac{3}{10}$

(B) Probability that A and B belongs to the committee is $\frac{1}{15}$

(C) Probability that either A and B belong to the committee is $\frac{8}{15}$

(D) Probability that exactly one of A and B belongs to the committee is $\frac{7}{15}$

Q4. A problem of mathematics is given to three students A, B and C. Whose chances of solving it are 1/2, 1/3 , 1/4 respectively. Then probability that the problem is solved is-

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) $\frac{3}{16}$

Q5. The probability that a student passes in Mathematics, Physics and Chemistry are m, p and c respectively. Of these subjects, the student has a 75% chance of passing in atleast one, a 50% chance of passing in atleast two, and a 40% chance of passing in exactly two, which of the following relations are true?

(A) $p+m+c=\frac{19}{20}$ (B) $p+m+c=\frac{27}{20}$

(C) $pmc=\frac{1}{10}$ (D) $pmc=\frac{1}{4}$

Q6. Two dice are thrown together 4 times. The probability that both dice will show same numbers exactly twice is

(A) $\frac{1}{3}$ (B) $\frac{25}{36}$

(C) $\frac{25}{216}$ (D) None

Q7. A manufacturer of airplane parts makes a certain engine that has a probability p of failing on any given flight. There are two planes fitted with this type of engine. One plane has 3 such engines and other plane has 5. A plane crashes if more than half the engines fitted in it fail. If the two plane models have the same probability of crashing then the value of p can be

(A) 0 (B) 1

(C) $\frac{1}{2}$ (D) $\frac{3}{5}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques.8 to 10)

Two bags A and B contains 2 white, 3 black, 4 red and 3 white, 4 black, 5 red balls resepectively. If one ball is drawn from bag A and put in the bag B. Now if one ball is drawn from bag B then :

Q8. The probability that drawn ball from bag B is white, is

(A) $\frac{49}{117}$ (B) $\frac{29}{117}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Q9. The probability that drawn ball from bag B is black, is

(A) $\frac{49}{117}$ (B) $\frac{29}{117}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Q10. The probability that drawn ball from bag B is red, is

(A) $\frac{49}{117}$ (B) $\frac{29}{117}$ (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Paragraph # 2 (Ques.11 to 13)

A box contains b red balls, '2b' white balls and '3b' blue balls where b is a positive integer. 3 balls are selected at random from the box.

Q11. If balls are drawn without replacement and 'A' denotes the event that "No two of the selected balls have the same colour" then

(A) there is no value of b for which $P(A) = 0.3$

(B) There is exactly one value of b for which $P(A) = 0.3$ and this value is less than 5

(C) There is exactly one value of b for which $P(A) = 0.3$ and this value is greater than 5

(D) There are two values of b for which $P(A) = 0.3$

Q12. If balls are drawn without replacement and 'B' denotes the event that "No two of the 3 drawn balls are blue" then

(A) $P(B) = \frac{1}{3}$ if $b = 1$

(B) $P(B) = \frac{2}{3}$ if $b = 2$

(C) $P(B) = \frac{1}{4}$ if $b = 4$

(D) $P(B) = \frac{1}{2}$ for all value of b

Q13. If $P(A) = 0.3$, then the value of $P(A/B)$ equals

(A) $\frac{3}{5}$ (B) $\frac{3}{10}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q14. The letters of the word PROBABILITY are written down at random in a row. Let E_1 denotes the event that two I′s are together and E_2 denotes the event that two B′s are together, then

Column-I	**Column-II**
(A) $P(E_1) = P(E_2)$ is equal to -	(P) $\frac{2}{55}$
(B) $P(E_1 \cap E_2)$ is equal to -	(Q) $\frac{2}{11}$
(C) $P(E_1 \cup E_2)$ is equal to -	(R) $\frac{1}{5}$
(D) $P(E_1/E_2)$ is equal to -	(S) $\frac{18}{55}$

Q15.

Column–I	Column–II
(A) The probability of hitting a target by three men is $\frac{1}{2}, \frac{1}{3}$ and $\frac{1}{4}$ respectively, if the probability that exactly two of them will hit the target is λ, then λ is	(P) 5/24
(B) A bag contains 4 white and 2 black balls, another bag contains 3 white and 5 black balls. one ball is drawn from each bag, then the probability that both are black balls is	(Q) 5/18
(C) Two balls are drawn form a bag containing 2 white, 3 red and 4 black balls one by one without replacement, then the probability that both the balls are of same colour is	(R) 1/4

SUBJECTIVE TYPE

Q16. Words are formed with the letters of the word PEACE. Find the probability that 2 E's come together.

Q17. A bag contains 5 red and 4 green balls. Four balls are drawn at random then find the probability that two balls are of red and two balls are of green colour.

Q18. Two dice are thrown at a time. Find the probability of the following

(i) these numbers shown are equal

(ii) the difference of numbers shown is 1.

Q19. Find the odds in favours of getting a king when a card is drawn from a well shuffled pack of 52 cards.

Q20. In a single cast with two dice find the odds against drawing 7.

Q21. There are four letters and four envelopes, the letters are placed into the envelopes at random, find the probability that all letters are placed in the wrong envelopes.

Q22. Two cards are drawn from a pack of 52 cards. What is the probability that either both are red or both are kings ?

Q23. A die is thrown twice and the sum of the numbers appearing is observed to be 6.

What is the conditional probability that the number 4 has appeared at least once?

Q24. A bag contains 3 red, 6 white and 7 blue balls. Two balls are drawn one by one. What is the probability that first ball is white and second ball is blue when first drawn ball is not replaced in the bag?

Q25. A bag contains 4 red and 4 blue balls. Four balls are drawn one by one from the bag, then find the probability that the drawn balls are in alternate colour.

Q26. Three coins are tossed together. What is the probability that first shows head, second shows tail and third shows head?

Q27. Let an ordinary fair dice is thrown for five times. If $P = \frac{a}{b}$ expressed in lowest form be the probability that the outcome of the fifth throw was already thrown, then find the value of $(a + b)$.

Q28. A bag contains N balls, some of which are white, the others are black, white being more in number than black. Two balls are drawn at random from the bag, without replacement. It is found that the probability that the two balls are of the same colour is the same as the probability that they are of different colour. It is given that $180 < N < 220$. If K denotes the number of white balls, find the exact value of $(K + N)$.

Q29. Let A, B, C be 3 independent events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$, $P(C) = \frac{1}{4}$.

Then find probability of exactly 2 events occuring out of 3 events.

Q30. A bag A contains 2 white and 3 red balls and a bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and it is found to be red. Find the probability that it was drawn from the bag B.

Q31. A covered basket of flowers has some lilies and roses. In search of rose, Sweety and Shweta alternately pick up a flower from the basket but puts it back if it is not a rose. Sweety is 3 times more likely to be the first one to pick a rose. If sweety begin this 'rose hunt' and if there are 60 lilies in the basket, find the number of roses in the basket.

Q32. There are 4 urns. The first urn contains 1 white & 1 black ball, the second urn contains 2 white & 3 black balls, the third urn contains 3 white & 5 black balls & the fourth urn contains 4 white & 7 black balls. The selection of each urn is not equally likely.

The probability of selecting i^{th} urn is $\frac{i^2+1}{34}$ $(i = 1, 2, 3, 4)$. If we randomly select one of the urns & draw a ball, then the probability of ball being white is p/q where p and $q \in N$ are in their lowest form. Find $(p + q)$.

Q33. Consider 4 independent trials in which an event A occurs with probability $\frac{1}{3}$. The event B will occur with probability 1 if the event A occurs atleast twice, it can not occur if the event A does not occur and it occurs with a probability $\frac{1}{2}$ if the event A occurs once. If the probability P of the occurrence of event B can be expressed as $\frac{m}{n}$, find the least value of $(m + n)$, where $m, n \in N$.

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If two of the 64 squares are chosen at random on a chess board, the probability that they have a side in common is -

(A) $\frac{1}{9}$ (B) $\frac{1}{18}$

(C) $\frac{2}{7}$ (D) none

Q2. Mr. A lives at origin on the cartesian plane and has his office at (4,5). His friend lives at (2,3) on the same plane. Mr. A can go to his office travelling one block at a time either in the +y or +x direction. If all possible paths are equally likely then the probability that Mr. A passed his friends house is -

(A) $\frac{1}{2}$ (B) $\frac{10}{21}$

(C) $\frac{1}{4}$ (D) $\frac{11}{21}$

Q3. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is -

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{2}{5}$ (D) $\frac{1}{5}$

Q4. For a biased die the probabilities for the different faces to turn up are given below :

Faces	1	2	3	4	5	6
Probabilities	0.10	0.32	0.21	0.15	0.05	0.17

The die is tossed & you are told that either face one or face two has turned up. Then the probability that it is face one is :

(A) $\frac{1}{6}$ (B) $\frac{1}{10}$

(C) $\frac{5}{49}$ (D) $\frac{5}{21}$

Q5. A is one of the 6 horses entered for a race and is to be ridden by one of two jockeys B or C. It is 2 to 1 that B rides A, in which case all the horses are equally likely to win, if C rides A, his chance is trebled, Then the odds against his winning are :

(A) 5 : 13 (B) 18 : 5

(C) 13 : 18 (D) 13 : 5

Q6. A card is drawn & replaced in an ordinary pack of 52 playing cards. Minimum number of times must a card be drawn so that the probability of getting heart is atleast 1/2.

(A) 2 (B) 3

(C) 4 (D) more than four

Q7. A die is rolled three times, the probability of getting a large number than the previous number is

(A) $\frac{5}{216}$ (B) $\frac{5}{54}$

(C) $\frac{1}{6}$ (D) $\frac{5}{36}$

Q8. If two events A and B are such that

$P(A^c) = 0.3.\ P(B) = 0.4$

and $P(A \cap B^c) = 0.5$,

then $P(B / A \cup B^c) =$

(A) 0.9 (B) 0.5

(C) 0.6 (D) 0.25

Q9. 7 persons are stopped on the road at random and asked about their birthdays. If the probability that 3 of them are born on Wednesday, 2 on Thursday and the remaining 2 on Sunday is $\frac{K}{7^6}$, then K is equal to

(A) 15 (B) 30

(C) 105 (D) 210

Q10. A & B are two independent events such that $P(\bar{A}) = 0.7, P(\bar{B}) = a$ & $P(A \cup B) = 0.8$, then a =

(A) $\frac{5}{7}$ (B) $\frac{2}{7}$

(C) 1 (D) none

Q11. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\bar{A}) = \frac{1}{4}$, where $\bar{A}$ stands for complement of event A. Then events A and B are

(A) mutually exclusive and independent

(B) independent but not equally likely

(C) equally likely but not independent

(D) equally likely and mutually exclusive

Q12. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is

(A) 2, 4, or 8 (B) 3, 6 or 9

(C) 4 or 8 (D) 5 or 10

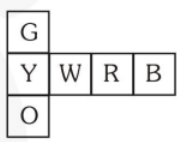
Q13. Three critics review a book. Odds in favour of the book are 5 : 2, 4 : 3 and 3 : 4 respectively for the three critics. The probability that majority are in favour of the book is

(A) $\frac{35}{49}$ (B) $\frac{125}{343}$ (C) $\frac{164}{343}$ (D) $\frac{209}{343}$

Q14. A box contains 24 balls of which 12 are black and 12 are white. The balls are drawn at random from the box one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is

(A) $\frac{5}{64}$ (B) $\frac{27}{32}$ (C) $\frac{5}{32}$ (D) $\frac{11}{32}$

Q15. A butterfly randomly lands on one of the six squares of the T-shaped figure shown and then randomly moves to an adjacent square. The probability that the butterfly ends up on the R square is



(A) $\frac{1}{4}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{1}{6}$

Q16. 'A' and 'B' each have a bag that contains one ball of each of the colours blue, green, orange, red and violet. 'A' randomly selects one ball from his bag and puts it into B's bag. 'B' then randomly selects one ball from his bag and puts it into A's bag. The probability that after this process the contents of the two bags are the same, is

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{5}$ (D) $\frac{1}{6}$

Q17. Two aeroplanes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. the probability that the targert is hit by the second plane is

(1) 7/50 (B) 7/22

(C) 7/10 (D) 7/100

Q18. An urn contains 2 white and 2 black balls. A ball is drawn at random If it is white, it is not repleaced into urn otherwise it is replaced alongwith another ball of the same colour. The process is repeated. The probability that the third ball drawn is black is

(A) $\frac{23}{30}$ (B) $\frac{3}{20}$

(C) $\frac{19}{60}$ (D) $\frac{1}{2}$

Q19. A box has four dice in it. Three of them are fair dice but the fourth one has the number five on all of its faces. A die is chosen at random from the box and is rolled three times and shows up the face five on all the three occasions. The chance that the die chosen was a rigged die, is -

(A) $\frac{216}{217}$ (B) $\frac{215}{219}$

(C) $\frac{216}{219}$ (D) none

Q20. A fair dice is thrown until 1 comes, then probability that 1 comes in even number of trials is -

(A) $\frac{5}{11}$ (B) $\frac{5}{6}$

(C) $\frac{6}{11}$ (D) $\frac{1}{6}$

Q21. Let E^c denote the complement of an event E. Let E,F,G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P(E^c \cap F^c | G)$ equals

(A) $P(E^c) + P(F^c)$ (B) $P(E^c) - P(F^c)$

(C) $P(E^c) - P(F)$ (D) $P(E) - P(F^c)$

Q22. An ant is situated at the vertex A of the triangle ABC. Every movement of the ant consists of moving to one of other two adjacent vertices from the vertex where it is situated. The probability of going to any of the other two adjacent vertices of the triangle is equal. The probability that at the end of the fourth movement the ant will be back to the vertex A, is -

(A) $\frac{4}{16}$ (B) $\frac{6}{16}$

(C) $\frac{7}{16}$ (D) $\frac{8}{16}$

Q23. The probablity that the length of a randomly chosen chord of a cirlce lies between $\frac{2}{3}$ and $\frac{5}{6}$ of its diameter if

(A) $\frac{5}{16}$ (B) $\frac{1}{16}$

(C) $\frac{1}{4}$ (D) $\frac{5}{12}$

Q24. If $a \in [-20, 0]$, then probability that the graph of the function $y = 16x^2 + 8(a+5)x - 7a - 5$ is strictly above the x-axis is

(A) $\frac{3}{20}$ (B) $\frac{13}{20}$

(C) $\frac{17}{20}$ (D) $\frac{1}{2}$

Exercise - 2A — Brain Booster

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Two whole numbers are randomly selected and multiplied. Consider two events E_1 and E_2 defined as

E_1 : Their product is divisible by 5

E_2 : Unit's place in their product is 5

Which of the following statement(s) is/are correct?

(A) E_1 is twice as likely to occur as E_2

(B) E_1 and E_2 are equiprobable

(C) $P(E_2/E_1) = 1/4$

(D) $P(E_1/E_2) = 1$

Q2. If E_1 and E_2 are two events such that $P(E_1) = 1/4$, $P(E_2/E_1) = 1/2$ and $P(E_1/E_2) = 1/4$

(A) then E_1 and E_2 are independent

(B) E_1 and E_2 are exhaustive

(C) E_2 is twice as likely to occur as E_1

(D) Probabilities of the events $E_1 \cap E_2$, E_1 and E_2 are in G.P

Q3. Two events A and B are such that the probability that at least one of them occurs is $\frac{5}{6}$ and both of them occurring simultaneously is $\frac{1}{3}$. If the probability of not occurrence of B is $\frac{1}{2}$ then

(A) A and B are equally likely

(B) A and B are independent

(C) $P(A/B) = \frac{2}{3}$

(D) $3\,P(A) = 4\,P(B)$

Q4. The probabilities of events, $A \cap B$, A, B & $A \cup B$ are respectively in A.P. with probability of second term equal to the common difference. Therefore the events A and B are

(A) mutually exclusive

(B) independent

(C) such that one of them must occur

(D) such that one is twice as likely as the other

Q5. A and B are two random events such that $0 < P(A) < 1$ and $0 < P(B) < 1$. Then which of the following are true?

(A) $P\left(\frac{A}{B}\right) > P(A) \Rightarrow P\left(\frac{B}{A}\right) > P(B)$

(B) $P\left(\frac{B}{A}\right) + P\left(\frac{B}{\overline{A}}\right) = 1$

(C) $P\left(\frac{A}{B}\right) = P\left(\frac{\overline{A}}{B}\right)$ only when $A \cap B = \phi$

(D) $P\left(\frac{B}{A}\right) = P\left(\frac{B}{\overline{A}}\right)$ only when A and B are independent

Q6. If A and B are two events such that $P(A) = 1/2$ and $P(B) = 2/3$, then

(A) $P(A \cup B) \geq \frac{2}{3}$

(B) $P(A \cap B') \leq \frac{1}{3}$

(C) $\frac{1}{6} \leq P(A \cap B) \leq \frac{1}{2}$

(D) $\frac{1}{6} \leq P(A' \cap B) \leq \frac{1}{2}$

Q7. For two events A and B, $P(A \cap B)$ is

(A) not less than $P(A) + P(B) - 1$

(B) not greater than $P(A) + P(B)$

(C) equal to $P(A) + P(B) - P(A \cup B)$

(D) equal to $P(A) + P(B) + P(A \cup B)$

Q8. Three numbers are chosen at random without replacement from $\{1,2,3,....10\}$. The probability that minimum of the chosen numbers is 3 or their maximum is 7, cannot exceed

(A) $\frac{11}{30}$ (B) $\frac{11}{40}$

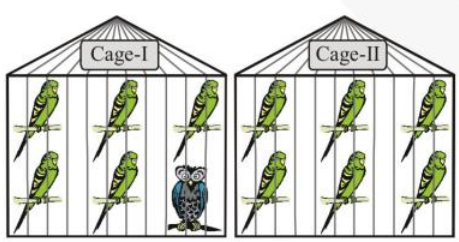
(C) $\frac{11}{50}$ (D) $\frac{11}{60}$

Q9. A coin is so weighted such that the probability of it showing H (Head) is $\frac{2}{3}$ and that of T(Tail) is $\frac{1}{3}$ when it is tossed. If head appears, then a number from the first 9 naturals is selected at random, otherwise a number from 1, 2, 3, 4, 5 will be selected. Let E be the event of getting an even number. Then,

(A) $P\left(\frac{E}{H}\right)=\frac{4}{9}$ (B) $P\left(\frac{E}{T}\right)=\frac{2}{5}$

(C) $P(E)=\frac{58}{135}$ (D) $P\left(\frac{H}{E}\right)=\frac{20}{29}$

Q10. Shalu bought two cages of birds : Cage-I contains 5 parrots and 1 owl, and Cage-II contains 6 parrots, as shown One day Shalu forgot to lock both cages and two birds flew from Cage-I to Cage-II. Then two birds flew back from Cage-II to Cage-I. Assume that all birds have equal chance of flying, the probability that the Owl is still in Cage-I, is



Birds like to fly

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Q11. There are three coins C_1, C_2 and C_3, C_1 is a fair coin painted blue on the head side and white on the tail side. C_2 and C_3 are biased coins so that the probability of a head is p. They are painted blue on the tail side and red on the head side. Two of the three coins are selected at random and tossed. If the probability that both the coins land up with sides of the same colour is $\frac{29}{96}$ then the possible value(s) of p can be

(A) $\frac{1}{8}$ (B) $\frac{3}{8}$ (C) $\frac{5}{8}$ (D) $\frac{7}{8}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 12 to 14)

Consider the independent events A, B, C corresponding to a random experiment. Suppose the event A × B × C represents the event of occurrence of atleast one of A, B, C and the event A.B. C represents the event of simultaneous occurrence of A, B, C. Also, the event $\overline{A}$ represents non-occurrence of A. It is given that : P(A) = a, P(A × B × C) = 1 –b, P(A. B. C.) = 1 –c and $P\left(\overline{A}.\overline{B}.C\right)=x$.

Now answer the following questions.

Q12. The probability of the event $\overline{A}.\overline{B}.\overline{C}$ is

(A) c (B) 1 – x

(C) b (D) 1 – a – b – c

Q13. The probability of occurrence of event B is

(A) $\frac{x}{x+b}$ (B) $\frac{(1-c)(x+b)}{ax}$

(C) $\frac{(1-a)^2+ab}{1-a}$ (D) $\frac{(1-b)(x-c)}{ax}$

Q14. The probability of occurrence of event C is

(A) $\frac{x}{x+b}$ (B) $\frac{(1-c)(x+b)}{ax}$

(C) $\frac{(1-a)^2+ab}{1-a}$ (D) $\frac{(1-b)(x-c)}{ax}$

Paragraph # 2 (Ques. 15 to 17)

2^n ($n \in N, n \geq 2$) players of equal strength are playing a knock out tournament.

They are paired randomly in all the rounds, and the winner reaches the next round.

Q15. The probability P_n that exactly one of the two specified players P_1 and P_2 reaches the semifinals is given by

(A) $\frac{8(2^n-4)}{2^n(2^n-1)}$ (B) $\frac{6(2^n-4)}{2^n(2^n-1)}$

(C) $\frac{4(2^n-4)}{(2^n-2)(2^n-1)}$ (D) $\frac{8(2^n-4)}{(2^n-2)(2^n-1)}$

Q16. If there are 16 players including P_1 and P_2 then the chance that exactly one of either P_1 or P_2 reaches the semifinals, is

(A) $\frac{8}{35}$ (B) $\frac{3}{10}$

(C) $\frac{2}{5}$ (D) $\frac{16}{25}$

Q17. As n tends to infinity then the value of P_n

(A) approaches $\frac{1}{2}$

(B) decreases and tends to approches zero

(C) equals zero

(D) first increases upto 0.5 and then decreases to zero value.

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18. An urn contain six red balls and four black balls. All ten balls are drawn from the urn, one by one and their colour is noted. Balls are not replaced once they have been drawn.

Column-I	**Column-II**
(A) Probability that first three balls are of same colour	(P) $\frac{1}{5}$
(B) Probability that last three balls are of same colour	(Q) $\frac{3}{5}$
(C) If it is known that first three are of the same colour, then the probability that colour is red is	(R) $\frac{4}{15}$
(D) Probability that no two consecutive balls in first three draw are of same colour is	(S) $\frac{5}{6}$

Q19. **Column -I** **Column - II**

(A) Two balls are drawn from an urn containing 2 white, 3 red and 4 black balls one by one without replacement. If the probability that both the balls are of same colour is λ and at least one ball is red is μ, then

(P) $\mu=\frac{8}{15}$

(B) A bag contains 2 white and 4 black balls while another bagcontains 6 white and 4 black balls, A bag is selected at random and a ball is drawn. If λ be the probability that the ball drawn is of white colour and μ be the probability that the ball drawn is black colour, then

(Q) $\mu=\frac{7}{12}$

(C) Bag A contains 4 red and 5 black balls and bag B contains 3 red and 7 black balls. One ball is drawn from bag A and two from bag B. If λ be the probability that out of 3 balls drawn two are black and one is red and μ be the probability that out of three balls drawn two are red and one is black, then

(R) $\lambda=\frac{5}{18}$

(S) $\lambda=\frac{7}{15}$

(T) $\mu=\frac{11}{45}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q20. In a multiple choice question there are four alternative answers of which one or more than one is correct. A candidate will get marks on the question only if he ticks all the correct answers. The candidate ticks the answers at random. If the probability of the candidate getting marks on the equation is to be greater than or equal to 1/3, then find the least number of chances he should be allowed.

SUBJECTIVE TYPE

Q21. A cube with all six faces coloured is cut into 64 cubical blocks of the same size which are thoroughly mixed. Find the probability that the 2 randomly chosen blocks have 2 coloured faces each.

Q22. Consider a function $f(x)$ that has zeroes 4 and 9. Given that Mr. A randomly selects a number from the set {–10, –9,–8,.....8, 9, 10}, what is the probability that Mr. A chooses a zero of $f(x^2)$?

Q23. Nine numbers 1,2,3,.....,9 are arranged in a rectangular array of matrix of order 3 so that each number occur exactly once. Find the probability that the sum of the numbers in atleast one horizontal row is greater than 21.

Q24. The probability that a person will get an electric contract is $2/5$ and the probability that he will not get plumbing contract is $4/7$. If the probability of getting at least one contract is $2/3$, what is the probability that he will get both?

Q25. A hotel packed breakfast for each of the three guests. Each breakfast should have consisted of three types of rolls, one each of nut, cheese and fruit rolls. The preparer wrapped each of the nine rolls and once warpped, the rolls were indistinguishable from one another. She then randomly put three rolls in a bag for each of the guests. If the probability that each guset got one roll of each type is m/n where m and n are relatively prime integers, find the value of $(m + n)$.

Q26. A person goes to office either by car, scooter, bus or train probability of which being $\frac{1}{7}$, $\frac{3}{7}$, $\frac{2}{7}$, and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}$, $\frac{1}{9}$, $\frac{4}{9}$, and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car.

Q27. During a power blackout, 100 persons are arrested on suspect of looting. Each is given a polygraph test. From past experience it is known that the polygraph is 90% reliable when administered to a guilty person and 98% reliable when given to some one who is innocent. Suppose that of the 100 persons taken into custody, only 12 were actually involved in any wrong doing. If the probability that a given suspect is innocent given that the polygraph says he is guilty is a/b where a and b are relatively prime, find the value of $(a + b)$.

Q28. Integers a, b, c and d not necessarily distinct, are chosen independently and at random from the set $S = \{0, 1, 2, 3, \ldots\ldots 2006, 2007\}$. If the probability that $|ad - bc|$ is even, is $\frac{p}{q}$ where p and q are relatively prime the find the value of $(p + q)$.

Q29. A match between two players A and B is won by the player who first wins two games. A's chance of winning, drawing or losing any particular games are 1/2, 1/6 or 1/3 respectively. If the probability of A's winning the match can be expressed in the form p/q, find $(p + q)$.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	D	D	A	B	B	D	C	D	C
11	12	13	14	15	16	17	18	19	20
A	D	B	A	A	C	A	B	B	A
21	22	23	24	25	26	27	28	29	30
A	C	C	A	A	C	B	C	C	C
31	32	33	34						
B	A	B	A						

Ex - 1A

1. B,D **2.** B,C,D **3.** A,B,C,D **4.** C **5.** B,C
6. C **7.** A,B,C **8.** B,C **9.** C **10.** A
11. B **12.** D **13.** A
14. (A) → (Q), (B) → (P), (C) → (S), (D) → (R)
15. (A) → (R), (B) → (P), (C) → (Q) **16.** $\frac{2}{5}$ **17.** $\frac{10}{21}$
18. (i) $\frac{1}{6}$ **(ii)** $\frac{5}{18}$ **19.** 1 : 12 **20.** 5 : 1 **21.** $\frac{3}{8}$ **22.** $\frac{55}{221}$
23. $\frac{2}{5}$ **24.** $\frac{7}{40}$ **25.** $\frac{6}{35}$ **26.** $\frac{1}{8}$ **27.** 1967
28. 301 **29.** $\frac{1}{4}$ **30.** $\frac{25}{52}$ **31.** 120 **32.** 2065
33. 130

Ex-2

1	2	3	4	5	6	7	8	9	10
B	B	C	D	D	B	B	D	B	B
11	12	13	14	15	16	17	18	19	20
B	D	D	C	A	B	A	A	C	A
21	22	23	24						
C	B	C	B						

EX - 2A

1. C,D **2.** A,D **3.** B,C,D **4.** A,D **5.** A,D
6. A,B,C,D **7.** A,B,C **8.** A,B **9.** A,B,C,D **10.** D
11. C,D **12.** C **13.** C **14.** A **15.** A
15. C **17.** B
18. (A) → (P), (B) → (P), (C) → (S), (D) → (R)
19. (A) → (Q,R), (B) → (P), (C) → (S, T), **20.** 5
21. $\frac{23}{168}$ **22.** $\frac{4}{21}$ **23.** $3 \cdot 3! \cdot 6!$ **24.** $\frac{17}{105}$ **25.** 79
26. $\frac{1}{7}$ **27.** 179 **28.**13 **29.** 206

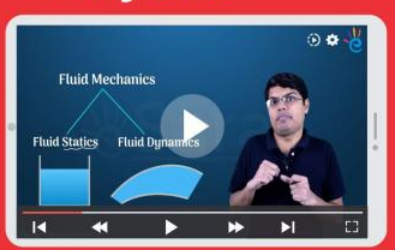
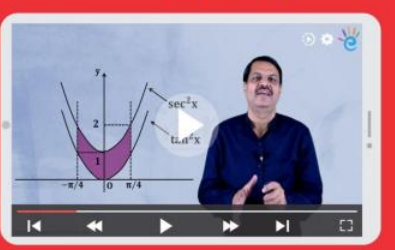
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